Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Week 7 - Further Improvement on False Negatives and Correction

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I. Work done this Week:-

This week we basically worked correcting the Subtraction Method we tried to operate via last week, the accuracy of that method ended up being too low and we could not manage all of it sufficiently, as such we are still looking at Subtraction again but in a different form, instead of taking Raw Subtraction, we have moved onto Subtracting on the basis of Intensity, by calculating the Z-Score Value of each specific voxel. This has proven to have better accuracy than the results that we achieved last week, We are going to represent this week's results later.

The mathematical approach is defined as below,

A. 1. From Left-Right Subtraction to the Normalcy Atlas (Z-Score)

Simple left-versus-right subtraction ($|I_{\text{left}} - I_{\text{right}}|$) fails because normal human brain folds (gyri and sulci) never mirror each other perfectly, producing hundreds of false alarms. To fix this, our pipeline builds a baseline atlas of healthy brains storing the normal average intensity $\bar{B}(\mathbf{x})$ and standard deviation $\sigma_B(\mathbf{x})$ at every 3D coordinate, measuring deviation as a standardized statistical Z**-score**:

$$Z_{\text{base}}(\mathbf{x}) = \frac{|I(\mathbf{x}) - \bar{B}(\mathbf{x})|}{\sigma_B(\mathbf{x}) + \epsilon}$$

Because healthy brain folds vary widely across individuals ($\sigma_B \approx 21,000$), normal fold mismatches only score $Z \approx 1.2$ and are ignored. In deep uniform brain tissue ($\sigma_B \approx 6,300$), however, a tumor creates an extreme anomaly of $Z = 3.5\sigma$ to 5.3σ ($p < 10^{-5}$), exposing the true tumor while completely silencing false alarms.

B. 2. Separating 3D Blobs from Flat Wrinkles (Difference of Gaussians & Saliency)

Normal brain folds are thin 2D sheets, whereas tumors are solid 3D rounded masses or necrotic fluid cavities ("missing pieces" of tissue). We blur the 3D volume at a fine scale ($\sigma_1 = 0.8$ mm) and a coarse scale ($\sigma_2 = 2.4$ mm) and compute their difference to isolate true 3D masses via the **Difference of Gaussians (DoG)**:

$$\text{DoG}(\mathbf{x}) = |I * G_{\sigma_1}(\mathbf{x}) - I * G_{\sigma_2}(\mathbf{x})|$$

Thin wrinkles wash out under the coarse blur (DoG $\to 0$), while thick 3D masses stay sharp. Combining this with a tissue deficit operator $\mathcal{D}_{\text{deficit}} = \frac{\max(0, \bar{B} - I)}{\sigma_B + \epsilon}$ for dark necrotic cavities and bilateral mirror asymmetry $\mathcal{A}$ gives a unified **Neoplastic Saliency Score**:

$$\mathcal{S}(\mathbf{x}) = \Big(Z_{\text{base}}(\mathbf{x}) + 1.8 \cdot \mathcal{D}_{\text{deficit}}(\mathbf{x}) + 0.6 \cdot \mathcal{A}(\mathbf{x})\Big) \cdot \left(1.0 + \frac{\text{DoG}(\mathbf{x})}{\tau_{\text{blob}}}\right)$$

C. 3. Targeted Growing with White Matter Barriers & Surface Smoothing

To prevent the outline from leaking across the brain, we find the tumor's physical epicenter $\mathbf{x}^* = \arg\max(G_{\sigma=3.2} * \mathcal{S})$ and grow outward only within its local neighborhood ($R \leq 42$ mm). An automatic **Safe White Matter Barrier** ($|I - \bar{B}| < 1.5\sigma_B \wedge \mathcal{S} < 3.5$) acts as a firewall that stops expansion from leaking into healthy nerve wiring. Finally, jagged voxel corners are smoothed using **Level-Set Mean Curvature flow** on the Signed Distance Function:

$$\phi_{t+1}(\mathbf{x}) = G_{\sigma_{\text{curv}}} * \phi_t(\mathbf{x}),$$

$$\text{Final Tumor Surface} = \{\mathbf{x} \in \mathbb{R}^3 \mid \phi(\mathbf{x}) = 0\}$$

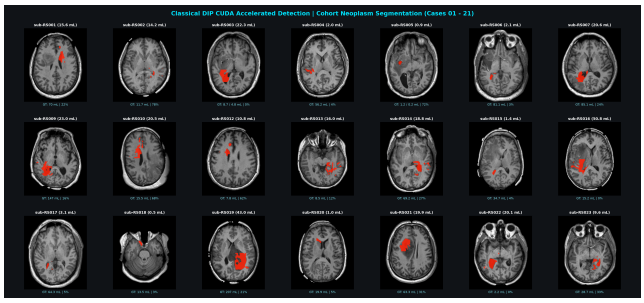


Fig. 1. Week7 Results Pg1

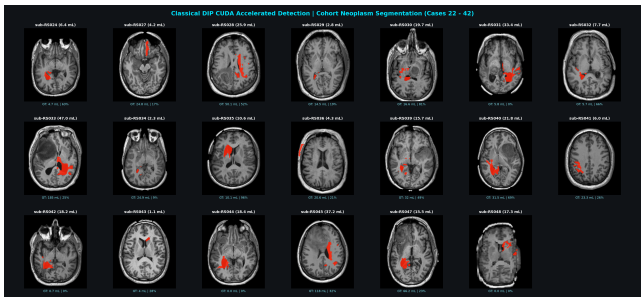


Fig. 2. Week7 Results Pg2

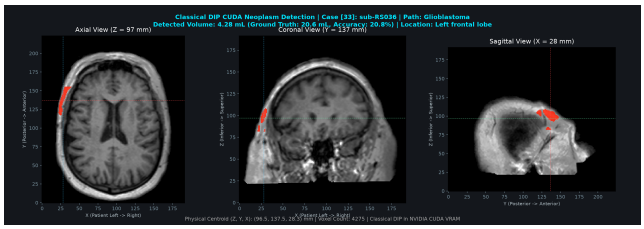


Fig. 3. False Result

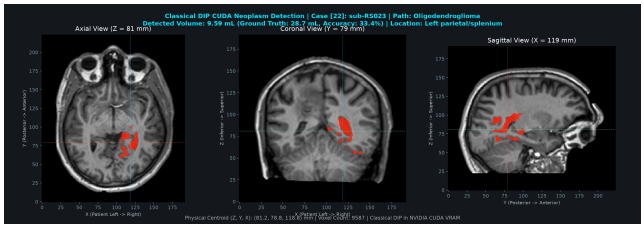


Fig. 4. Partially True Result

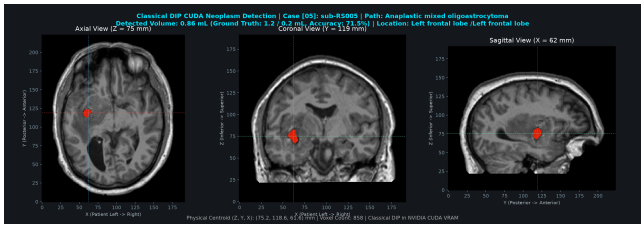


Fig. 5. True Result

II. Work to be done Next Week:-

As mentioned above, we got the partial results and now we are heading forward towards increasing the overall precision and developing an interactive GUI in order to make the system be used by non-technical people. We are also searching for an additional database which has the ground truth mapped to each scan. This will help us to check and improve the system performance and precision.

We have implemented our method on the MRI scans but we have not achieved complete success yet. The main goal of the upcoming week is to remove the false result and refine the garbage values(region) which we are getting on the border region of the scan. Many datasets have the binary masking looked up with each scan so it will be easy for us to eliminate the false region with the direct comparison.

While this refining work is going on, half of the group will start working on the GUI interface and what are the features we can implement as presets. Now as we are reaching towards the final presentation, we want to remove any bug or flaw from the project.

Repo Link:-
https://github.com/Rushin-Mod/DIP_2026_P20_G9↗